

The quadratic formula

HANDOUT



Interactive version

GOAL

In this lesson

BY THE END OF THE LESSON

- ☐ I can prove the quadratic formula by completing the square, explaining why each step is allowed.
- ☐ I can solve a quadratic equation with integer coefficients by the formula when D is a perfect square, and check a root by substituting it.
- ☐ I can solve a quadratic equation whose roots are irrational: simplify $\sqrt{D}$ and write the exact roots.

2 class periods • about 90 min

WHAT YOU NEED FROM BEFORE

- ↔ Lesson 1.1 • Quadratic equations: what we can already solve
- ↔ Lesson 1.2 • Solving by completing the square
- ↔ Square roots

HOOK

Two gardens

A rectangular garden is 3 m longer than it is wide. Its area is 40 m^2 .

Level 1 • threshold

How wide is the garden?

Now a second garden, also 3 m longer than it is wide, has area 30 m^2 .

Level 3 • basic

How wide is this garden? Round to 0.01 m .

EXPLORE

Where the parabola meets the axis

Solving $ax^2 + bx + c = 0$ means finding the x -coordinates of the points where the parabola $y = ax^2 + bx + c$ meets the x -axis. The parabola can cross the axis, touch it, or miss it. Move the sliders and watch the number $D = b^2 - 4ac$ under the figure.

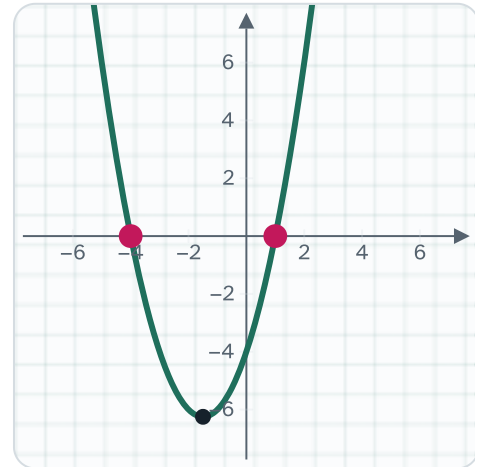
Level 2 · satisfactory

Predict first. Keep $a = 1$ and $b = 3$. Which value of c makes the parabola touch the x -axis at exactly one point? Then move c to check.

Level 1 · threshold

When the parabola does not meet the x -axis at all, what is D ?

- ☐ A negative ☐ B positive ☐ C zero



$D = b^2 - 4ac = 25$, so two roots: $x_1 \approx -4$, $x_2 \approx 1$.

EXPLAIN

Why the formula works

A square and a strip

Take $x^2 + 6x$. Draw x^2 as a square with side x , and $6x$ as a strip that is 6 wide and x long.

Split the strip

Cut the strip into two strips, each $3 \times x$. Put one beside the square and one below it. The area is still $x^2 + 6x$.

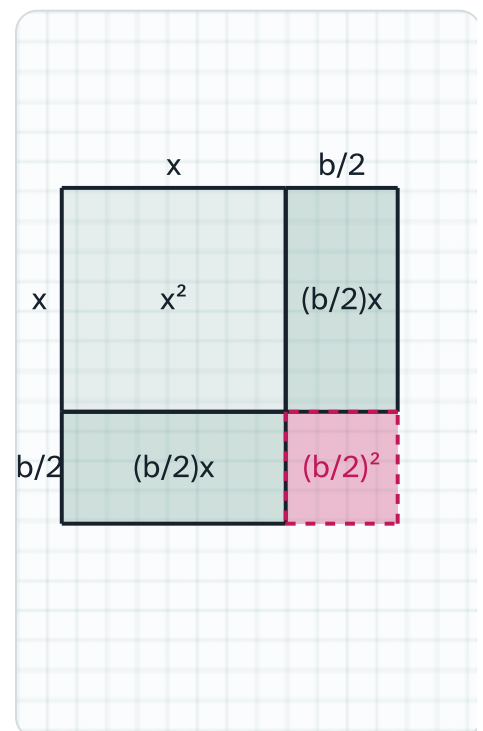
Complete the square

One 3×3 corner is missing. Adding 9 completes a square with side $x + 3$:

$$x^2 + 6x + 9 = (x + 3)^2.$$

The picture shows the case $x > 0$; the identity holds for every real x .

Adding 9 to both sides of $x^2 + 6x = 7$ gives $(x + 3)^2 = 16$, so $x + 3 = \pm 4$: $x = 1$ or $x = -7$.



Do it for any equation

The same moves work for $ax^2 + bx + c = 0$ with $a \neq 0$. The column on the right says what is done to both sides to get the next line.

$ax^2 + bx + c = 0$	$: a$
$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$	$-\frac{c}{a}$
$x^2 + \frac{b}{a}x = -\frac{c}{a}$	$+\left(\frac{b}{2a}\right)^2$
$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a}$	left side is a square
$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$	

The formula

Call $D = b^2 - 4ac$ the **discriminant**. The right side is $\frac{D}{4a^2}$, and $4a^2 > 0$.

- If $D > 0$: take square roots, $x + \frac{b}{2a} = \pm \frac{\sqrt{D}}{2a}$. Two roots:

$$x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}.$$

- If $D = 0$: one root, $x = -\frac{b}{2a}$.
- If $D < 0$: a square cannot be negative, so there are no real roots.

WORKED EXAMPLES

Three worked examples

Example 1 • Two roots: $2x^2 - 3x - 2 = 0$

COEFFICIENTS

$$a = 2, b = -3, c = -2.$$

DISCRIMINANT

$$D = (-3)^2 - 4 \cdot 2 \cdot (-2) = 9 + 16 = 25 > 0, \text{ so there are two roots.}$$

FORMULA

$$x_{1,2} = \frac{3 \pm 5}{4}, \text{ so } x_1 = 2 \text{ and } x_2 = -\frac{1}{2}.$$

CHECK

$$2 \cdot 2^2 - 3 \cdot 2 - 2 = 8 - 6 - 2 = 0. \checkmark$$

Example 2 • The second garden: $x^2 + 3x - 30 = 0$

COEFFICIENTS

$$a = 1, b = 3, c = -30.$$

DISCRIMINANT

$$D = 3^2 - 4 \cdot 1 \cdot (-30) = 9 + 120 = 129 > 0.$$

FORMULA

$$x_{1,2} = \frac{-3 \pm \sqrt{129}}{2}, \text{ so } x_1 \approx 4.18 \text{ and } x_2 \approx -7.18.$$

ANSWER

A width cannot be negative, so the garden is $\frac{-3 + \sqrt{129}}{2} \approx 4.18$ m wide.

Example 3 • One root: $4x^2 - 4x + 1 = 0$

COEFFICIENTS

$$a = 4, b = -4, c = 1.$$

DISCRIMINANT

$$D = (-4)^2 - 4 \cdot 4 \cdot 1 = 16 - 16 = 0, \text{ so there is one root.}$$

ROOT

$$x = -\frac{b}{2a} = \frac{4}{8} = \frac{1}{2}.$$

CHECK

$$4x^2 - 4x + 1 = (2x - 1)^2, \text{ which is 0 only at } x = \frac{1}{2}. \checkmark$$

SUMMARY

What to remember

- Write the equation as $ax^2 + bx + c = 0$ first, then read a, b, c **with their signs**.
- The discriminant $D = b^2 - 4ac$ tells you how many roots there are:
 - $D > 0$: two roots;
 - $D = 0$: one root, $x = -\frac{b}{2a}$;
 - $D < 0$: no real roots.
- The roots are $x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$. The formula comes from completing the square.
- Check a root by substituting it. In a word problem, keep only the roots that make sense.